

Lakshya JEE 2027

KPP - 01

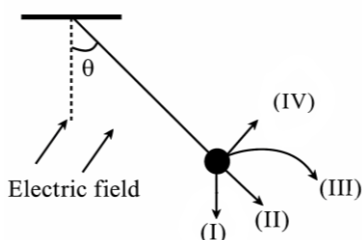
Electrostatics

By: Saleem Bhaiya

1. Two point charges $q_1 = 2 \mu\text{C}$ and $q_2 = -2 \mu\text{C}$ are located at $x = 0, y = 0.3 \text{ m}$ and $x = 0, y = -0.3 \text{ m}$ respectively. The net electric force on a third point charge $Q = 4 \mu\text{C}$ at $x = 0.4 \text{ m}, y = 0$ is:

- (1) $(0.46\text{N})\hat{i}$ (2) $(0.34\text{N})-\hat{j}$
(3) $(0.34\text{N})\hat{j}$ (4) $(-0.46\text{N})\hat{i}$

2. A small object of mass M and charge Q is connected to an insulating massless string in vacuum on Earth. A uniform electric field exists throughout the region of the vacuum as indicated. The mass remains in static equilibrium at an angle θ with the vertical as shown in the figure. When the string is cut, which of the illustrated paths best indicates the trajectory of the mass?



- (1) I (2) II
(3) III (4) IV

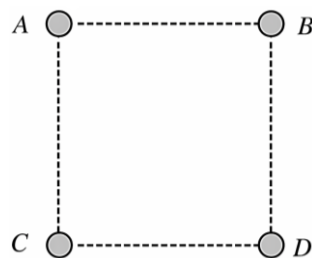
3. Two identical charges, $5 \mu\text{C}$ each, are in the air. What is the largest charge to be transferred from one charge to another, so that their interaction force decreases 2.5 times?

- (1) $\sqrt{11} \mu\text{C}$ (2) $4 \mu\text{C}$
(3) $\sqrt{13} \mu\text{C}$ (4) $\sqrt{15} \mu\text{C}$

4. A point charge $-3q$ is positioned at point $(2, -1, 3)$ of a Cartesian coordinate system. A test point charge $2q$ is placed at point $(3, 0, 4)$. Which of the following is the unit vector in the direction of the force acting on the test charge? All the coordinates are in meters.

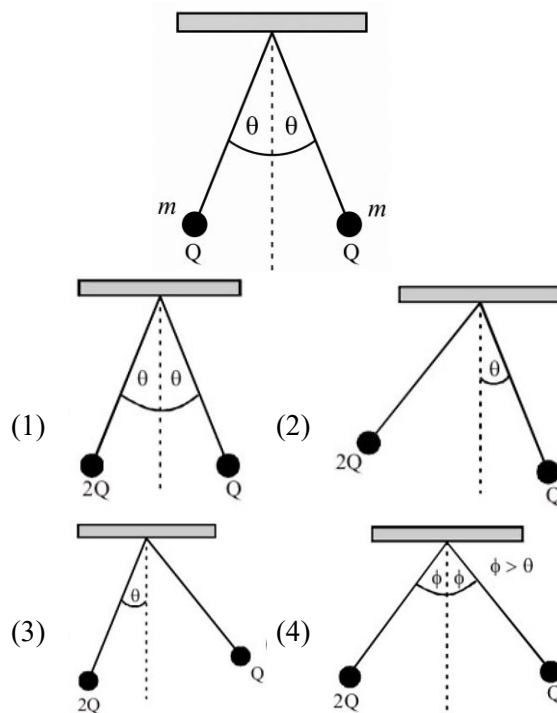
- (1) $\hat{i} + \hat{j} + \hat{k}$ (2) $-(\hat{i} + \hat{j} + \hat{k})$
(3) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ (4) $-\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$

5. Four charges are at the corners of a square as shown in the figure. Corners A and D have equal charge, while both B and C have a charge of $+1 \text{ C}$. What is the charge on A so that the force on B is zero?



- (1) -1.0 C (2) -0.5 C
(3) -0.35 C (4) -0.71 C

6. Two pith balls with mass m are suspended from insulating threads. When the pith balls are given equal positive charge Q , they hang in equilibrium as shown. We now increase the charge on the left pith ball from Q to $2Q$ while leaving its mass essentially unchanged. Which of the following diagrams best represents the new equilibrium configuration?



7. Two charges, each equal to q , are kept at $x = -a$ and $x = a$ on the x -axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to:

- (1) y (2) $-y$
(3) $\frac{1}{y}$ (4) $-\frac{1}{y}$

8. Two identical charged spheres suspended from a common point by two massless strings of length ℓ are initially a distance d ($d \ll \ell$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result, the charges approach each other with a velocity v . Then as a function of distance x between them:

- (1) $v \propto x^{1/2}$ (2) $v \propto x$
(3) $v \propto x^{-1/2}$ (4) $v \propto x^{-1}$

9. Given are four arrangements of three fixed electric charges. In each arrangement, a point labeled P is also identified — test charge, $+q$, is placed at point P. All of the charges are of the same magnitude, Q , but they can be either positive or negative as indicated. The charges and point P lie on a straight line. The distances between adjacent items, either between two charges or between a charge and point P, are all the same. Correct order of choices in a decreasing order of magnitude of force on P is:



- (1) $\text{II} > \text{I} > \text{III} > \text{IV}$ (2) $\text{I} > \text{II} > \text{III} > \text{IV}$
(3) $\text{II} > \text{I} > \text{IV} > \text{III}$ (4) $\text{III} > \text{IV} > \text{I} > \text{II}$

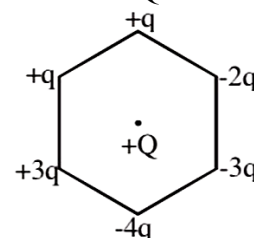
10. Two point charges of $100 \mu\text{C}$ and $-4 \mu\text{C}$ are positioned at points $(-2\sqrt{3}, 3\sqrt{3}, -4)$ and $(4\sqrt{3}, -5\sqrt{3}, 6)$ respectively of a Cartesian coordinate system. Find the force vector on the $-4 \mu\text{C}$ charge? All the coordinates are in meters.

- (1) $9 \times 10^{-4} (3\sqrt{3} \hat{i} - 4\sqrt{3} \hat{j} + 5 \hat{k})$
(2) $9 \times 10^{-4} (-3\sqrt{3} \hat{i} + 4\sqrt{3} \hat{j} - 5 \hat{k})$
(3) $2.25 \times 10^{-4} (-3\sqrt{3} \hat{i} + 4\sqrt{3} \hat{j} - 5 \hat{k})$
(4) $2.25 \times 10^{-4} (3\sqrt{3} \hat{i} - 4\sqrt{3} \hat{j} + 5 \hat{k})$

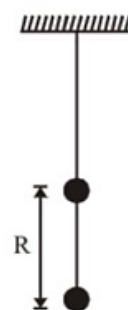
11. Two charged balls are connected by an inextensible thread of length 3 m. Masses of balls are 2 kg and 1 kg, the charges are $+20 \mu\text{C}$ and $-100 \mu\text{C}$. What minimum constant external force F (in N) must be applied to the ball of mass 1 kg so that the thread does not slack? Neglect gravity and friction.

12. Four identical charges q are placed at the corners of a square of side a , what charges Q must be placed at the centre of the square so that whole system of charges is in equilibrium?

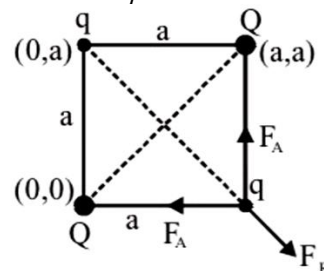
13. Six charges are kept at the vertices of a regular hexagon as shown in the figure. If magnitude of force applied by $+Q$ on $+q$ charge is F , then net electric force on the $+Q$ is nF . Find the value of n .



14. Two identical balls of mass $m = 0.9 \text{ g}$ each are charged by the same charges, joined by a thread and suspended from the ceiling (figure). Find the charge (in μC) that each ball should have so that the tension in both the threads are same? The distance between the centers of balls is $R = 3 \text{ m}$.



15. For the system shown in figure find Q for which resultant force on q is zero.





Answer Key

1. (2)

2. (2)

3. (4)

4. (4)

5. (3)

6. (4)

7. (1)

8. (3)

9. (3)

10. (2)

11. (3)

12. $\frac{(1+2\sqrt{2})}{4Q}$

13. (9)

14. (003)

15. $q = -2\sqrt{2} Q$